LIGATURE: Generative Inference of Spatial Molecular Organization from Histology

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Abstract

Virtual stains predict molecular profiles from hematoxylin and eosin (H&E) images, but their per-cell objectives do not target the relationships underlying molecular clustering and immune co-localization. Given cell segmentations, we formulate spatial molecular inference as learning residual dependence targeted to downstream statistics. Conditional-mean plug-in estimates of bilinear statistics omit cross-cell residual covariance; even exact per-cell marginals leave their expectations unidentified, with semidefinite bounds for Gaussian marginals. LIGATURE implements this formulation around a frozen virtual stain, using morphology-conditioned multiscale Gaussian fields and region- and patient-level latent variables to generate joint molecular samples. Strictly proper scores on region- and patient-level spatial statistics align training with downstream analyses. In a reference patient-disjoint evaluation of 40 colorectal cancer patients with paired H&E and 16-plex immunofluorescence, median attenuation of non-negligible cohort means decreased from 34% to 6%. Clustering mean squared error in training-standardized coordinates fell by 40% relative to the stain (39% over four runs). In an exploratory comparison under independent calibration across three seeds, overall interval score was 10.6% below a matched direct graph's at approximately 90% coverage; clustering-interval and point-error comparisons were inconclusive. Frozen samples supported gated contacts excluded from training, with 30% lower error than a quantile-matched stain, although adapted predictors performed better on new autocorrelation queries. In a separate 19-patient lung Xenium evaluation, overall and clustering error decreased by 16% and 52% relative to its stain. These results support dependence modeling for spatial inference, with gains dependent on the query and intensity scale; interval calibration requires separate molecular measurements.

1 Introduction

Spatial tissue analyses quantify how molecular states vary across neighboring cells: immune infiltration of tumor nests, macrophage–checkpoint contacts, and clustering of proliferating or lineage-defined cells. Inferring these quantities from routine hematoxylin and eosin (H&E) requires predictions for region- and patient-level statistics. Analyses may introduce new gates, neighborhoods and distances after training. Virtual multiplexing models already predict molecular markers from H&E at population scale [1–3]; GigaTIME, for example, generated virtual immunofluorescence for 14,256 patients [1]. We target spatial-statistic estimates and probabilistic forecasts from these cell predictions.

We introduce LIGATURE, a statistic-scored generative residual model. It keeps a frozen virtual stain as its mean and fits an image-conditioned distribution over the residuals of all cells jointly. A cell-graph transformer reads the tissue and parameterizes loadings on spatial Gaussian fields, cell noise, and region and patient latents. Training scores cell markers, spatial moments and patient summaries with the continuous ranked probability score (CRPS), so spatial dependence is fitted against the analyses it must support. At inference, the same joint samples supply estimates and predictive distributions for both the scored statistics and additional nonlinear queries. Query-specific calibration on separate measured patients provides marginal prediction intervals.

The statistical rationale connects this design to the target quantities. For a bilinear statistic, the gap between its conditional expectation and the plug-in of a per-cell-optimal predictor is a weighted residual covariance (Proposition 1). Per-cell marginal laws leave this covariance unidentified; we characterize the resulting range by certified semidefinite computations (Proposition 2). Scoring spatial statistics instead specifies their conditional-law targets under a correctly specified generator class (Propositions 3–4), while likelihood uses different covariance weights under misspecification (Proposition 7). Generated maps are realizations of the fitted model; our validation concerns their spatial-statistic forecasts, without claiming recovery of an individual tissue's unique unobserved molecular configuration.

Our contributions are: (i) a joint residual generator and analysis-directed training procedure that connect frozen virtual stains to spatial inference, with shared sampling across regions and patient-disjoint statistic-level calibration; (ii) statistical rationale and computable identification bounds for the dependence targeted by the procedure; and (iii) a patient-held-out evaluation across six stains, two cohorts and assay modalities, direct and probabilistic baselines, systematic ablations, and nonlinear statistics absent from training.

2 Related work

Virtual multiplex from histology. ROSIE [2], HEX [3], GigaTIME [1], generative image translators [4] and diffusion-based multimodal histopathology models such as MuPD [5] predict protein maps from H&E; related models represent measured panels (VirTues [6]), impute missing markers (CytoVI [7]) or choose markers to measure (ECHO-k [8]). Per-cell/pixel and downstream evaluations do not establish the spatial-statistic law. Concurrently, CellPlex [9] benchmarks per-cell protein prediction from H&E across seven cohorts, including ORION-CRC; our target is the spatial-statistic law computed from molecular predictions, rather than per-cell prediction accuracy.

Uncertainty for imputed spatial omics. TISSUE [10] gives conformal per-cell intervals for imputed spatial transcriptomes and propagates them to downstream tests by multiple imputation, and SpatialNP [11] applies neural processes to probabilistic multiplexed tissue imaging. Per-unit marginals alone do not specify cross-cell statistics (Proposition 2). Bounds on dependence given marginals are classical (Fréchet–Hoeffding bounds, statistical matching); the proposition specializes the Gaussian-marginal case to spatial statistics and makes it computable and certified at tissue scale.

Prediction-powered and design-based correction. Prediction-powered inference [12, 13], semi-supervised U-statistics [14], and spatially dependent PPI [15] combine model predictions with a labeled subsample for valid inference on population-level targets. LIGATURE predicts for individual patients conditional on supplied cell geometry; its predictions can serve as inputs to such estimators.

Proper scoring rules and generative distributional regression. Energy-score training of noise-injected generators (engression [16]) and CRPS-trained generative weather ensembles [17] learn conditional distributions without likelihoods. We bring this principle to tissue and score the statistics analyzed down-

stream; as in weather forecasting [18, 19], a single high-dimensional energy score appeared insufficient in a preliminary run (Appendix A2).

3 Method

Task and interface. Training uses paired H&E and molecular measurements on segmented regions. Inputs at deployment are H&E cell features, cell positions and shape descriptors from a supplied segmentation, and a frozen stain; outputs are joint molecular draws and statistics computed on them. Primary evaluations use assay segmentations; a separate H&E-derived proxy control changes both geometry and target definition (Appendix A5). Interval calibration for new queries uses their values on a separate measured patient set. This evaluates statistical inference conditional on the supplied segmentation.

A region (crop) of a section contains cells $i = 1, \dots, n$ at positions x_i with context H collecting H&E features and supplied geometry, and molecular vector $Y_i \in \mathbb{R}^p$ (standardized log intensities). Write $d_{ij} = \|x_i - x_j\|$ and let $|E_w|$ denote the pair-count normalization for radial weight w. For markers (a, b), a bilinear spatial statistic is

$$Q_{w,ab}(Y) = \frac{1}{|E_w|} \sum_{i \neq j} w(d_{ij}) \, \tfrac{1}{2} \, (Y_{ia} Y_{jb} + Y_{ib} Y_{ja}). \tag{1}$$

Our primary contrast uses $w = \mathbf{1}\{d \leq 25\,\mu\text{m}\}$ normalized by its pair count minus $\mathbf{1}\{50 < d \leq 75\,\mu\text{m}\}$ normalized likewise. It compares nearby marker products with a distant within-region reference; for $a = b$ it measures same-marker clustering in training-standardized log intensities. Centering and scaling define the estimand. Original-log-intensity contrasts and Moran's I/Geary's C provide complementary evaluations (Section 6.5). Patient-level statistics average crops with equal weight.

Definition 1 (Spatial statistic vector)**.** For distance bins $B_1, \dots, B_K$, let $S(Y)$ collect the cell means, the binned symmetric second moments $G_{k,ab} = |E_k|^{-1} \sum_{i \neq j:\, d_{ij} \in B_k} \frac{1}{2}(Y_{ia} Y_{jb} + Y_{ib} Y_{ja})$ for every bin k and unordered pair $a \leq b$, and the bin contrasts $G_{k,ab} - G_{k',ab}$.

Generative model. LIGATURE wraps a frozen per-cell virtual stain $f(H) \in \mathbb{R}^{n \times p}$ with an image-conditioned joint residual distribution. A model draw is

$$Y_i = f_i(H) + \sum_{\ell=1}^{4} A_\ell(H)_i \, \xi_{\ell, r(i)}(x_i) + B(H)_i \, \epsilon_i + C(H)_i \, \zeta_{r(i)} + D(H)_i \, \eta_g, \tag{2}$$

where $r(i)$ indexes cell i's region, and $\xi_{\ell,r}$ are independent four-variate unit Gaussian random fields with squared-exponential correlation at length scales $\ell \in \{10, 20, 40, 80\}\,\mu$m (approximated with 64 random Fourier features per field and region), ϵ_i is cell noise, ζ_r a region latent and η_g a patient latent shared by all regions of patient g. The loading matrices $A_\ell, B, C, D \in \mathbb{R}^{p \times \cdot}$ are per-cell outputs of a cell-graph transformer, so that the within-patient residual covariance $\mathrm{Cov}(Y_i, Y_j \mid H) = \mathbf{1}_{r(i)=r(j)}[\sum_\ell k_\ell(d_{ij}) A_{\ell i} A_{\ell j}^\top + C_i C_j^\top] + \mathbf{1}_{i=j} B_i B_i^\top + D_i D_j^\top$ is positive semidefinite by construction, cross-marker, spatially structured, and morphology dependent (a linear model of coregionalization whose coefficients are read from the tissue). If $f = \mu$, Proposition 1 makes the plug-in bias of every bilinear statistic the corresponding weighted sum of these covariances. Write $X_{k,ab}(f, \delta) = G_{k,ab}(f + \delta) - G_{k,ab}(f) - G_{k,ab}(\delta)$ for the bilinear cross term. For mean error $\delta = \mu - f$, suppressing bin/marker subscripts, matching a binned statistic's mean gives $\Gamma(\hat{\Sigma}) = \Gamma(\Sigma) + G(\delta) + X(f, \delta)$: the generated covariance absorbs the stain's mean error and equals Σ there only when $f = \mu$.

Architecture. Each cell is represented by a frozen pathology foundation-model embedding of its 20.8 μm H&E patch (Phikon-v2 [20], 1024-d, PCA to 128-d fitted on training patients), four nuclear and cell-shape descriptors, and the virtual-stain prediction f_i. Three transformer layers attend across all cells of a 166 μm region with a learned distance-dependent attention bias (16 radial basis functions per head). The primary ORION and lung stains use this transformer trained with per-cell squared error; each wrapper is compared with its own frozen stain's plug-in.

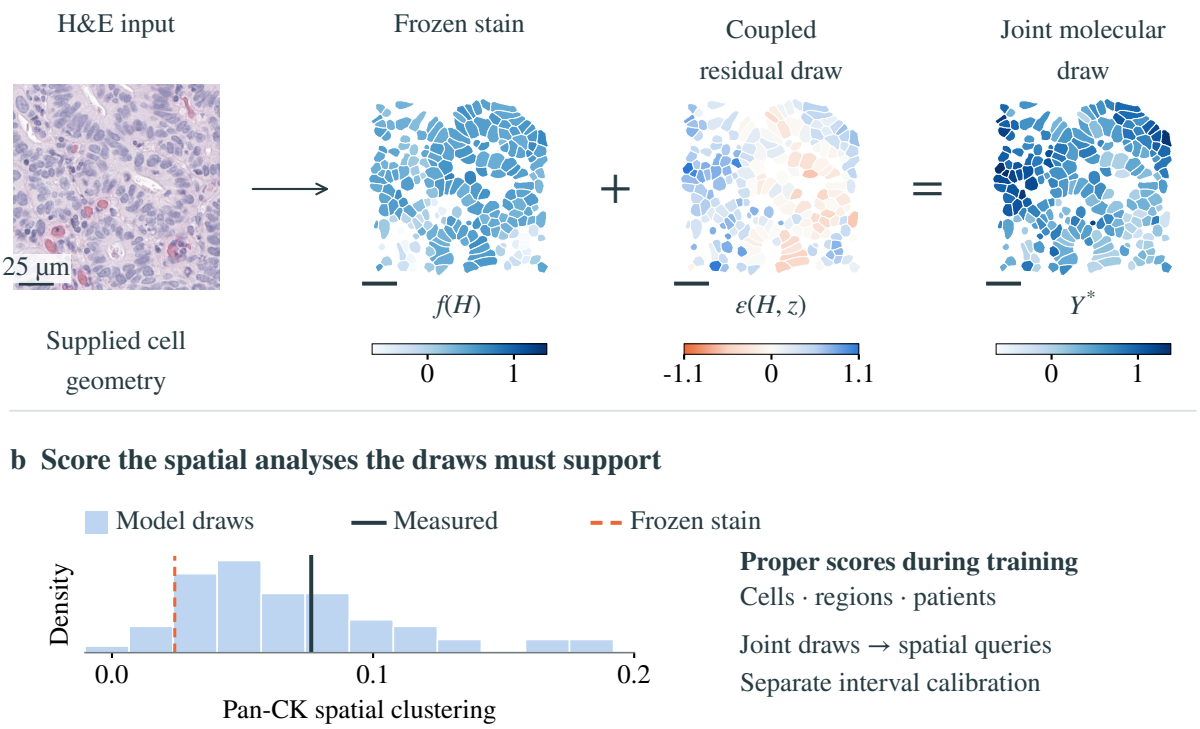


Figure 1: LIGATURE generates molecular values jointly in tissue space. **a**, Original H&E from the selected crop and molecular maps on its 196 retained, assay-segmented cells. The frozen Pan-CK prediction plus a coupled residual gives a joint draw. Views share physical coordinates and image orientation; bars: 25 μm. Molecular maps share a standardized-log-intensity scale; residuals use a signed scale. **b**, All 64 stored clustering draws, with measured and frozen-stain values marked. Training scores cells, regions and patients; intervals require separate calibration. This selected illustration does not establish aggregate accuracy or coverage. The molecular realization is separate from the histogram draws.

Three-level objective. For a batch of 4 patients with 4 regions each and $s = 1, \ldots, 16$ joint samples per region,

$$\mathcal{L} = \underbrace{\overline{\mathrm{CRPS}}\big(\mathcal{N}(f_{ia}, \sigma_{ia}^2), Y_{ia}\big)}_{\text{cell markers}} + \lambda_r \underbrace{\overline{\mathrm{CRPS}}\big(\{\theta_j(S(\hat{Y}_r^{(s)}))\}, \theta_j(S(Y_r))\big)}_{\text{region statistics}} + \lambda_p \underbrace{\overline{\mathrm{CRPS}}\big(\{\theta_j(\bar{S}_g^{(s)})\}, \theta_j(\bar{S}_g)\big)}_{\text{patient statistics}}, \qquad (3)$$

where the cell term scores a moment-matched Gaussian surrogate by closed-form CRPS, with σ_{ia}^2 the model's marginal variance, the other terms use the unbiased 16-sample CRPS over all $J = 832$ coordinates and bin contrasts of S (16 means, 3×136 binned moments, 3×136 contrasts) standardized by their standard deviation across training regions, and $\bar{S}_g$ averages S over the 4 sampled regions of patient g with samples sharing η_g. We use bins 0–25, 25–50 and 50–75 μm, $\lambda_r = \lambda_p = 1$, AdamW (learning rate 10^{-3} one-cycle, weight decay 10^{-3}, dropout 0.1), 15 epochs, and early stopping on validation patients. Hyperparameters were fixed before test evaluation. Only the linear residual is resampled, so an epoch over 24 patients takes about 30 s on 8 CPU cores.

Held-out variance calibration. A flexible model fit to 24 patients learns their statistics too sharply. After training we freeze the network and fit seven scalars, one log-scale for each variance component (four fields, cell, region, patient), on the validation patients with the same objective. Every downstream statistic inherits this rescaled spread, though raw intervals remain under-covered.

Inference. For a new slide we draw 64 joint samples (sample k of every region of a patient shares $\eta_{g,k}$). Applying a query to every sample gives its model-based predictive distribution and mean estimate. Held-

out query values calibrate its intervals; empirical sample quantiles alone do not guarantee coverage. For gated queries, validation-fitted quantile thresholds (LIGATURE-Q) adjust predicted positive fractions, using the same information as a quantile-matched plug-in. The resulting cell labels are marker-based proxies for phenotypes. Bilinear conditional means can also be integrated analytically (Appendix A2); nonlinear queries use samples.

4 Statistical rationale and identification limits

Proposition 1 (Plug-in bias). *Let $\mu = \mathbb{E}[Y \mid H]$ and $\Sigma_{ia,jb} = \mathrm{Cov}(Y_{ia}, Y_{jb} \mid H)$. For any predictor $f(H)$,*

$$\mathbb{E}[Q_{w,ab}(Y) \mid H] - Q_{w,ab}(f) = \big[Q_{w,ab}(\mu) - Q_{w,ab}(f)\big] + \frac{1}{|E_w|} \sum_{i \neq j} w(d_{ij}) \, \tfrac{1}{2} \, (\Sigma_{ia,jb} + \Sigma_{ib,ja}).$$

The proof expands $Y = \mu + (Y - \mu)$. For the per-cell-optimal $f = \mu$ the bias is the distance-weighted conditional cross-cell covariance of the molecular residual, the part of molecular organization the image does not reveal; shrinkage adds an attenuation term. Per-cell marginal calibration alone does not determine the cross-cell covariance term.

Proposition 2 (Sharp identification from per-cell laws). *Fix marker a, a symmetric weight matrix W with zero diagonal, and per-cell predictive laws $Y_{ia} \mid H \sim \mathcal{N}(\mu_i, \sigma_i^2)$. The set of values of $\mathbb{E}[Q(Y) \mid H] = \mathbb{E}[\sum_{i \neq j} W_{ij} Y_{ia} Y_{ja} \mid H]$ over all joint laws with these per-cell marginals is the interval*

$$Q(\mu) + \big[\min_{R \in \mathcal{E}_n} \langle \Omega, R \rangle, \ \max_{R \in \mathcal{E}_n} \langle \Omega, R \rangle \big], \qquad \Omega_{ij} = W_{ij} \sigma_i \sigma_j,$$

where $\mathcal{E}_n = \{R \succeq 0, \ R_{ii} = 1\}$ is the elliptope. Every value in the interval is attained by a Gaussian joint law.

Proof. $\mathbb{E}[Q \mid H] = Q(\mu) + \sum_{i \neq j} W_{ij} \sigma_i \sigma_j \, \mathrm{Corr}(Y_i, Y_j \mid H)$. Every correlation matrix lies in $\mathcal{E}_n$, and every $R \in \mathcal{E}_n$ is the correlation matrix of the Gaussian law $\mathcal{N}(\mu, \mathrm{diag}(\sigma) R \, \mathrm{diag}(\sigma))$, which has the prescribed marginals. The image of the convex set $\mathcal{E}_n$ under a linear map is an interval. □

The same argument covers cross-marker statistics with bivariate Gaussian per-cell laws (Appendix), again giving an interval with semidefinite endpoints. The endpoints are max-cut-type semidefinite programs, solved with a dual certificate (Appendix).

Proposition 3 (Scoring the statistics that will be analyzed). *(i) Every $Q_{w,ab}$ whose weight is constant on each bin is a fixed linear functional of S. (ii) Let $\theta_1, \ldots, \theta_J$ be linear functionals of S and let* CRPS *denote the continuous ranked probability score, which is strictly proper for univariate laws with finite mean [21]. The expected score $\sum_j \mathbb{E}\,\mathrm{CRPS}(F_{\theta_j}, \theta_j(S))$ is minimized exactly by predictive laws whose marginals along every θ_j equal the true conditional laws of $\theta_j(S) \mid H$. Hence a population minimizer in a generator class containing the truth reproduces the conditional mean and every quantile of each scored statistic.*

Proposition 4 (What statistic-level scores identify). *Let a Gaussian residual model around a frozen f have covariance $\Sigma(H)$, with bin averages $\Gamma_{k,ab}(\Sigma) = |E_k|^{-1} \sum_{d_{ij} \in B_k} \frac{1}{2}(\Sigma_{ia,jb} + \Sigma_{ib,ja})$. (i) If the class contains a law whose scored marginals are the true ones, every minimizer of the score of Proposition 3 satisfies $\Gamma_{k,ab}(\hat{\Sigma}) = \mathbb{E}[G_{k,ab}(Y) \mid H] - G_{k,ab}(f)$ for all k, a, b, so every bin-constant bilinear statistic, scored or not, has the correct conditional mean. (ii) The means of all scored statistics are invariant to every perturbation of Σ with zero bin averages, which includes all within-cell cross-marker covariances; these determine the conditional law of one marker given another in the same cell, which the scores therefore do not identify through their means.*

Part (ii) is consistent with the failure of conditioning on anchor markers (Section 7).

Gaussian likelihood and statistic scores weight covariance errors differently under misspecification (Appendix Proposition 7). In ORION regions only 9.2% of cell pairs lie within 25 μm, so a likelihood fit can attenuate the nearest-bin covariance as much as the plug-in does. The per-cell score is also strictly proper, but only for the per-cell law; by Proposition 2 its minimizers need not reproduce the law of any $\theta_j(S)$. A multivariate energy score on all of S is strictly proper for the joint law, but it is known to be weakly sensitive

to dependence and to individual coordinates in high dimension [18]; a preliminary generator trained this way left attenuation unchanged (Appendix A2). We therefore score each analyzed statistic directly.

Sampling a patient's regions independently understates the variance of their average by the design effect $1 + (m - 1)\bar{\rho}$ for average cross-region correlation $\bar{\rho}$ (Appendix Proposition 5). A statistic-averaged split-conformal construction (Appendix Proposition 6), a special case of conformal risk control [22], turns the generator's per-statistic means and spreads into patient-level intervals whose expected coverage fraction over a new patient's statistics is at least $1 - \alpha$ when the calibration patients are exchangeable and unseen by the model.

5 Experiments

Data. ORION-CRC [23] provides same-section H&E and 19-channel cyclic immunofluorescence with supplied whole-cell segmentation, distributed as 512×512-pixel regions at 0.325 μm/pixel. We use all 3,337 crops in the released VitaminP-derived ORION subset, whose source regions were selected for reliable segmentation; this is not a whole-slide sample. Quality control removes tiny/border cells, unmatched nuclei and duplicate cells in overlapping crops (Appendix). The retained dataset has 333,133 cells in 3,331 regions from 40 patients; 3,323 regions contain both near and far cell pairs. Targets are 16 biological markers (CD31, CD45, CD68, CD4, FOXP3, CD8a, CD45RO, CD20, PD-L1, CD3e, CD163, E-cadherin, PD-1, Ki67, Pan-CK, SMA); Hoechst, autofluorescence and the blank channel are excluded. Intensities are $\log(1 + x)$ of the mean whole-cell signal, standardized with training-patient statistics only.

Protocol. Five rotations assign 24 training, 8 validation and 8 test patients so that every patient is a test patient exactly once. Validation patients are used for early stopping, for LIGATURE's variance calibration and to fit the calibration baselines; no test-patient protein enters any fitted quantity. Contrast evaluation uses regions with both near and far pairs. We evaluate all 136 marker pairs (120 cross-marker, 16 same-marker) with the near (0–25 μm) minus far (50–75 μm) contrast at the patient level. Pooled summaries use each fold's training-standardized coordinates, rather than common assay units. Errors are normalized per statistic by the plug-in's patient-level mean squared error (MSE) and then averaged. Relative bias is the cohort-mean error divided by the measured mean, summarized over statistics with $|\text{measured mean}| > 0.02$. Reported p-values are exploratory and unadjusted for multiplicity; patient-bootstrap intervals quantify sampling uncertainty conditional on the fitted models, not model-selection uncertainty.

The reference run, reseed mean, four-run aggregate and rerotation are distinguished in Appendix Table 21; independent calibration uses separate patient roles.

Baselines. The plug-in uses the frozen stain's per-cell predictions (test cell MSE 0.78). Global rescaling fits one slope shared across statistics on validation regions. The matched direct graph receives the same embeddings and stain predictions, uses the same transformer, and predicts Gaussian laws for the scored statistic vector with region- and patient-level CRPS. It predicts statistics rather than cell measurements.

Additional controls are per-statistic affine calibration on validation regions; direct ridge regression without stain inputs, from region feature means/standard deviations, cell count and binned products of the leading 16 components, with its penalty chosen on validation patients; and independent region draws from LIGATURE, removing cross-region dependence. Three simple baselines were added post hoc, after test results were seen (Appendix A5.1). Ablations are in Section 6.6.

6 Results

6.1 Spatial-statistic estimation on held-out patients

The four-run aggregate of LIGATURE has relative MSE 0.91 overall and 0.61 for clustering; the direct graph (stain inputs), reseed mean, has 0.88 and 0.64 (Appendix Table 6). The direct model has lower overall error; the clustering difference is not statistically significant. The evidence favoring LIGATURE over direct prediction concerns calibrated overall forecasts; reusing samples avoids a new fit, without an established accuracy or cost advantage.

The reference run reduces median cohort attenuation from 34% to 6% overall and 58% to 5% for clustering (Appendix Fig. 3a). Relative MSE is 0.88 overall (patient-bootstrap 95% interval 0.833–1.002) and 0.60 for clustering (0.54–0.73). Clustering error is lower in 36 of 40 patients ($p = 5 \times 10^{-7}$); the overall per-patient

Table 1: Patient-level estimation of 136 spatial statistics over 40 held-out patients (24/8/8 training/validation/test rotations). MSE is relative to the plug-in (lower is better); bias is the median relative cohort-mean error; ρ is Spearman correlation with measured values across patients; CRPS treats point estimates as point masses. †Post hoc; one validation-fitted shift per statistic.

	MSE / plug-in			bias			
Method	all	cross	same	all	same	ρ	CRPS
Plug-in	1.00	1.00	1.00	−34%	−58%	0.52	0.040
Per-statistic calibration	1.13	1.17	0.78	0%	0%	0.41	0.042
Plug-in + constant†	0.98	1.02	0.69	+2%	+2%	0.44	–
Direct ridge regression	1.42	1.48	0.99	–	–	0.31	–
Direct graph (stain inputs)	0.89	0.93	0.63	–	–	0.53	–
Global rescale of plug-in	0.89	0.90	0.83	−18%	−49%	0.54	–
Calibrated Gaussian plug-in	1.00	1.00	1.00	−34%	−58%	0.52	0.0326
LIGATURE (reference run)	0.88	0.92	0.60	−6%	−5%	0.53	0.0274
LIGATURE (four-run aggregate)	0.91	0.95	0.61	−6%	−4%	0.52	0.0277

Except for the four-run aggregate, rows use the reference evaluation. Gaussian plug-in: per-statistic validation variance (90% coverage 86%, width 0.19). Reference LIGATURE coverage: 74.2% raw, 90.9% after validation inflation (width 0.18); independently sampled regions: 49.4%. The four-run aggregate row averages per-run metrics, not predictions; the reference run has the lowest overall MSE of those four.

improvement is not statistically significant ($p = 0.09$, 24 of 40 patients). Global rescaling is competitive overall (0.89) and slightly better on cross-marker statistics (0.90 vs 0.92), but leaves −18% median bias and higher clustering MSE (0.83 vs 0.60). One validation-fitted constant per statistic removes more of the cohort bias (median +2%), with MSE 0.98 overall and 0.69 for clustering, so bias correction accounts for most of the clustering gain; LIGATURE's MSE difference from it is −0.10 overall ($[-0.25, -0.02]$) and a borderline −0.09 for clustering ($[-0.46, 0.002]$), with better patient ranking (0.53 vs 0.44).

Reference-run CRPS is 0.0274 versus 0.0326 for a calibrated Gaussian plug-in: 16.1% lower. It is lower than the point-mass plug-in's 0.040 in all 40 patients ($p = 2 \times 10^{-12}$). Mean patient-ranking correlation changes little overall (0.52 to 0.53), with a larger change for clustering (0.55 to 0.60); cohort bias correction and patient discrimination are distinct outcomes.

6.2 Patient-level uncertainty

Independent calibration separates selection from calibration: stain and generator are retrained on 20 patients per rotation, 2 serve selection/variance fitting, and 10 serve calibration only. We extend this protocol to three fixed residual/direct-model seeds, reusing the five stains. All 40 patients are tested once per seed; metrics are averaged across seeds, not ensembles. Each method receives the same per-statistic split-conformal calibration. Adaptive intervals scale with predicted spread; constant-scale controls use the same predicted center (Fig. 2; Table 2).

Across all 136 statistics, LIGATURE's adaptive interval score is 10.6% below the matched direct graph's (difference −0.59, paired patient-bootstrap 95% interval $[-0.87, -0.31]$), with coverage 90.5% versus 90.9%. Widths are nearly equal (3.72 versus 3.75); the difference comes from smaller miss penalties (1.26 versus 1.82). Its score is also 12.2% below its own constant-scale control (difference −0.69, interval $[-0.98, -0.41]$), indicating information in the spread beyond the center. These overall comparisons are exploratory and largely driven by cross-marker forecasts. The prespecified clustering score difference from the direct graph is not statistically significant (−0.50, interval $[-1.11, 0.11]$), and the width-difference interval also includes zero. Point-MSE ratios are 0.95 for LIGATURE and 0.96 for direct prediction; the difference is not statistically significant. Raw sample quantiles cover only 70.2% overall and 44.8% for clustering. These development-cohort checks are not independent confirmation (Appendix A5).

Post hoc baselines do not close the overall gap but are not separated on clustering (Table 2; Appendix A5.1). LIGATURE's difference from the plug-in plus one constant is −0.60 ($[-0.91, -0.27]$; clustering −0.23, $[-0.93, 0.45]$), a baseline the direct graph only ties (−0.01, $[-0.21, 0.22]$); a variant using in-sample plug-ins has lower point error (MSE 0.93, clustering 0.68) at score 5.57. The difference from a stain-context

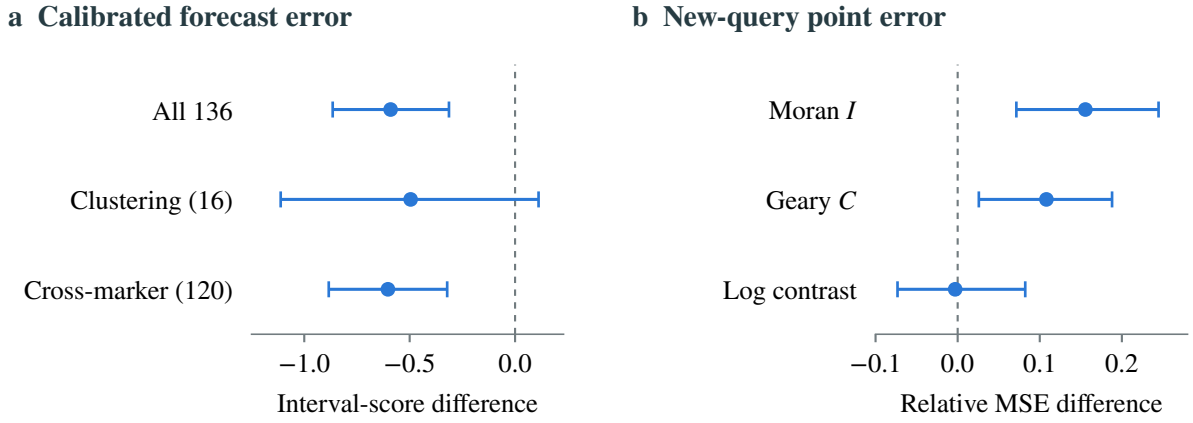


Figure 2: Complete three-seed comparison under independent calibration, all 40 patients. Differences are LIGATURE minus the matched direct graph: negative favors LIGATURE. **a**, Training-standard-deviation-normalized 90% interval score; overall coverage is 90.5% versus 90.9%. **b**, Plug-in-normalized point MSE for queries excluded from generative training; the direct head is trained on these queries. Points average seed metrics; bars are conditional 95% patient-bootstrap intervals. The overall forecast comparison is exploratory; clustering includes zero, while the new autocorrelation queries favor direct prediction.

ridge is -0.82 ($[-1.13, -0.47]$). That ridge's penalty selection on two patients is erratic: fixed penalties from 10 to 10^4 reach MSE 0.93–0.94 (clustering 0.63–0.68), at or below LIGATURE's 0.95 (0.66), with no penalty scoring below 5.61 overall.

The reference pipeline is exploratory because its validation patients also serve selection and variance fitting. Raw 90% intervals cover 74.2% overall (Appendix Fig. 3c), versus 49.4% with independently sampled regions. Validation inflation yields 90.9% (86–96% across rotations), but clustering coverage remains 50% raw and 76% inflated. Statistic-averaged conformal intervals with $\alpha = 0.15$ cover 97% overall and 91% for clustering; validation reuse prevents an exact finite-sample guarantee for this protocol. Its plug-in control also covers (92%), with clustering intervals 1.8 times wider.

Calibrating held-out values of unscored queries gives conformal 90% intervals for gated contact fractions and neighborhood frequencies that cover 91.6% and 92.8% (raw: 46% and 44%). They are about as wide as identically calibrated quantile-matched plug-in intervals, with 26% and 24% lower patient-averaged squared error (Table 16); 25% of adaptive contact intervals span all of $[0, 1]$.

6.3 Six virtual stains, including two published models

We wrapped five further stains without changing LIGATURE (Appendix Table 8): a per-cell ridge regression on the same embeddings (no spatial context); a pixel-level U-Net trained on H&E at 0.65 μm/pixel with pixel and per-cell losses, the architecture class of published image-to-image stains; two published models run on ORION H&E without fine-tuning; and one of them, GigaTIME, fine-tuned end to end on training patients. ROSIE [2] covers all 16 of our markers and GigaTIME [1] 9; both are mapped to ORION intensities on training patients (Appendix A2).

Attenuation depends strongly on the stain: -68% and -94% for GigaTIME and ROSIE under the domain shift from their training platforms, -34 to -39% for the cell-level stains, and -12% and -7% for the U-Net and fine-tuned GigaTIME. LIGATURE brings the median bias to within 8% for every stain. Relative to each stain's own plug-in, point-estimate gains track the attenuation: patient-level MSE is 0.61 and 0.73 for ROSIE and GigaTIME, 0.86–0.88 for the cell-level stains, and 1.04 for the U-Net and 0.96 for fine-tuned GigaTIME. Probabilistic accuracy improves for every stain (CRPS 10–34% below a calibrated Gaussian plug-in), and for gated contact fractions LIGATURE has lower error than the quantile-matched plug-in for every stain (0.40–0.72 vs 0.63–1.71 of the plug-in's).

Table 2: Independent calibration (20/2/10/8 roles), three seeds and 40 test patients. Direct denotes the graph with stain inputs. MSE is relative to the protocol's plug-in. Coverage, width and 90% interval score summarize all 136 statistics; width and score use training-patient standard deviations (lower is better). All intervals use the same ten calibration patients per rotation. †Post hoc (Appendix A5.1).

Method / interval	MSE / plug-in all	same	Coverage	Width	Score
Plug-in / constant	1.00	1.00	91.5%	4.09	5.89
Global rescale / constant	1.85	1.07	90.6%	4.88	7.13
No-image mean† / constant	1.63	1.34	89.8%	4.62	6.78
Plug-in + constant† / constant	1.01	0.73	90.9%	3.89	5.58
Stain-context ridge† / constant	1.03	0.71	90.9%	4.01	5.80
Direct / adaptive	0.96	0.75	90.9%	3.75	5.57
LIGATURE / adaptive	0.95	0.66	90.5%	3.72	4.98
LIGATURE / constant	0.95	0.66	91.1%	3.99	5.67

A common scale. Dividing every estimator's per-statistic MSE by the transformer plug-in's (Table 18) shows what LIGATURE takes from a stain. In the reference run, predicting the trained statistics directly is competitive: a graph model with the same embeddings, transformer and CRPS scores has 1.30 times the plug-in's error without the stain, and 0.89 (0.63 for clustering) with the stain's predictions as inputs, against 0.88 (0.60) for LIGATURE (lung: 0.82 vs 0.84; CRPS within 5% either way). Scoring the statistics matters too: the same backbone fitted by a regularized conditional Gaussian likelihood has 2.7 times the plug-in's error, worse than the attenuation Proposition 7 describes. LIGATURE around a constant stain reaches 1.13 (1.00 with a learned mean; 0.76 and 0.75 for clustering), and the published stains applied across platforms add nothing to it (wrapped ROSIE 1.17, GigaTIME 1.14), so their gains over their own plug-ins measure how much LIGATURE repairs a poorly transferred stain.

Why pixel-level stains look unbiased. They omit as much residual second moment as the other in-domain stains, but a negative predictor–residual cross term, which vanishes for a conditional mean, offsets it in the cohort mean (Appendix Table 14); a frozen-mean wrapper cannot undo it.

6.4 A separate cohort, platform and modality

We applied the identical pipeline, with no hyperparameter changed, to the 20 lung Xenium samples of HEST-1k [24], from 19 patients of a pulmonary fibrosis study [25] (15 fibrotic, 5 control samples): same-section H&E with in situ transcript counts, so tissue, disease, platform, molecular modality and segmentation all differ from ORION. We built 1,600 regions of 166 μm (216,102 cells) and took 16 genes spanning epithelial, immune, stromal and endothelial lineages (Appendix), with library-normalized counts as targets and patient rotations as before (11 training, 4 validation, 4 test). Results are in Table 3.

The plug-in attenuates the statistics by a median of 71%. LIGATURE lowers patient-level MSE to 0.84 of the plug-in's overall ($p = 0.005$; 13 of 19 patients) and 0.48 for clustering ($p = 3 \times 10^{-5}$; 17 of 19), and its CRPS is 23% below the calibrated Gaussian plug-in's in all 19 patients ($p = 4 \times 10^{-6}$). A global rescale gains nothing (1.03), the direct graph model reaches 0.93 (0.82 with the stain as input) and LIGATURE around a constant stain 0.90–0.91, so here the stain adds to the encoder. For gated contact fractions LIGATURE-Q's error is 46% below the quantile-matched plug-in's, again in all 19 patients ($p = 4 \times 10^{-6}$).

Attenuation is only partly removed (median bias −23%; one validation-fitted constant per statistic gives 0%, with MSE 0.96 overall and 0.64 for clustering), and raw 90% intervals cover 64% (89% after inflation); a patient-disjoint refit with 4 calibration patients, enough only for an 80% target, covers 82.6% of clustering statistics with intervals 19.5% narrower than the calibrated plug-in's, though a constant-scale interval around LIGATURE's mean is narrower still at a similar score (Table 17).

6.5 Statistics never scored in training

Joint samples support nonlinear neighborhood-maximum queries excluded from training. For all 256 ordered marker pairs, we estimate the fraction of a-positive cells with a b-positive neighbor within 20 μm, pooling counts over each patient's regions (at least 20 measured a-positive cells). Training-patient 85th-percentile gates define measured positives; marker gates are phenotype proxies. The plug-in underestimates these

Table 3: Separate lung cohort (Xenium, 19 held-out patients): patient-level MSE relative to the plug-in over 136 gene-pair statistics (all) and the 16 clustering statistics (same), mean Spearman correlation with measured values, and median relative bias of cohort means.

Estimator	MSE all	MSE same	ρ	bias
Plug-in (cell-graph transformer stain)	1.00	1.00	0.46	−71%
Global rescale of the plug-in	1.03	0.83	0.45	−60%
Direct graph model (statistics, no cells)	0.93	0.55	0.39	−21%
LIGATURE around a constant stain	0.91	0.52	0.43	−18%
LIGATURE around a constant stain, learned mean	0.90	0.53	0.47	−24%
Direct graph with stain inputs (statistics, no cells)	0.82	0.49	0.49	−24%
LIGATURE around the transformer stain	0.84	0.48	0.48	−23%

fractions (median bias −26%). Its validation-quantile gates target nominal 15% positivity, reduce bias to −6% and lower MSE to 0.88 of the raw plug-in's. LIGATURE reaches 0.62 (bias +9%), with lower error than the quantile-matched plug-in for 83% of pairs and 29 of 40 patients (paired Wilcoxon $p = 9 \times 10^{-5}$; ratio 0.70, patient-bootstrap 90% interval 0.65–0.83), and raises mean rank correlation from 0.50 to 0.59 (Table 12). Raw 90% coverage is 52%.

With validation-quantile thresholds (LIGATURE-Q), contact error falls to 0.56 and is below the quantile-matched plug-in's for 97% of pairs. A contact-trained direct graph (three seeds; Appendix A2) reaches 0.59 (lung 0.15 vs 0.17); paired 95% difference intervals are −0.07 to 0.06 and 0.00 to 0.05. ORION family-average errors are not statistically separated. An exploratory CD8a–epithelial proximity case favors LIGATURE-Q by 26% in MSE over that graph (95% ratio interval 0.559–0.916; Appendix A5.3); survival comparisons remain inconclusive.

The same holds for per-patient cellular neighborhood frequencies [26] (eight gated types; 10 neighborhoods of 10-nearest-neighbor compositions, fitted on training patients). Unmatched LIGATURE does not beat the quantile-matched plug-in (total-variation distance to the measured profile 0.36 vs 0.34), because its Gaussian residual over-calls positive cells; LIGATURE-Q does (0.29, Wilcoxon $p = 0.005$; per-neighborhood MSE 0.53 vs 1.17 of the plug-in's, clusters aligned across rotations), and does so for every stain in Appendix Table 8 (0.28–0.36 vs 0.33–0.54, $p \leq 0.006$ each; Table 13), including the U-Net. A Gaussian copula (training on per-marker normal scores) removes the need for matched thresholds (unmatched: neighborhood distance 0.30, $p = 5 \times 10^{-4}$; contact error 0.59). Other gates and three further query families behave similarly (Appendix A3.1).

Limits of query reuse. In the three-seed independent protocol, frozen samples improve Moran's I and Geary's C over the plug-in (MSE ratios 0.43 and 0.46), but an adapted direct graph is better (0.27 and 0.35; paired Holm-adjusted $p = 0.003$ and 0.009), as is a cheap frozen-feature ridge (0.21 and 0.29). Post hoc, a no-image training mean reaches 0.42 and 0.52 and the plug-in plus one constant 0.31 and 0.38, and both have better interval scores than LIGATURE's adaptive intervals (Moran 5.08 and 5.88 vs 7.89; Geary 5.35 and 4.86 vs 6.32), so the gain over the plug-in does not show added image information. Preserving the exact within-cell marginal law while removing cross-cell dependence gives 2.24 and 3.16: samples fail on these ratios without cross-cell dependence, but dependence does not make them accurate. Inverting fold standardization before clustering evaluation gives much smaller improvements on the original log-assay scale (sample mean 0.98, analytic mean 0.96 of the plug-in's MSE). The larger standardized-contrast gains are estimand dependent.

Downstream checks. Neither association-study error (0.099 vs 0.149 for the plug-in; 90% interval for the reduction −2% to 40%) nor predictive-spread triage (50% of error removed by measuring 20% of patients, vs 51% by plug-in magnitude) establishes additional biological utility (Appendix Fig. 5b,c).

6.6 Ablations

Across 36 ablation variants, statistic-level scores, bin contrasts, morphology-dependent loadings and shared patient variation matter most (Appendix Tables 4,5). Removing spatial fields raises overall MSE to 1.00 (lung 0.94 vs 0.84–0.86) but leaves ORION clustering (0.62) and CRPS (0.0282) within seed variation; one field scale or rank 1 also stays within noise. The reseed mean has MSE 0.92 ± 0.02, with raw coverage

$72.9 \pm 0.9\%$. In the rerotation, with the stain retrained, LIGATURE has MSE 0.93 overall and 0.64 for clustering, with median bias reduced from -34% to -5%; the direct graph (stain inputs) has 0.91 and 0.69. Cell-only training gives MSE 1.33; removing the patient latent gives raw coverage 49.2%. Earlier-run variants are labeled separately. Independent Gaussian per-cell residuals with fitted marginal variances leave attenuation unchanged and cover 28%, consistent with Proposition 2.

6.7 Dependence targeted by the generator

For model-implied Gaussian per-cell marginals, the certified identification interval for expected clustering is a median 2.4 times the statistic's 5–95% cohort range: per-cell laws leave this much dependence unspecified (Appendix A3.2).

Known-law simulations. We simulate residuals from the model class on ORION geometry with the frozen stain as the true conditional mean (Table 19). The Bayes per-cell plug-in still attenuates contrasts by 38% (clustering 61%). LIGATURE removes attenuation (median bias +1%), with MSE ratios 0.59 overall and 0.12 for clustering, versus 0.32 and 0.07 for the oracle statistic predictor. Near-bin cross-marker residual correlations agree with truth ($r = 0.97$). Added within-cell factors with correlations 0.22 and 0.36 remain unrecovered (0.01 and -0.20) with contrast MSE 0.61 overall and 0.12 for clustering, consistent with Proposition 4(ii).

Non-touching-pair sensitivity. A fixed-model check excludes near pairs whose supplied masks are within two pixels in Chebyshev distance ($0.65\,\mu$m per axis), removing 20.3% of near pairs. On 3,322 retained regions and all 40 patients, clustering MSE is 0.72 of the matched plug-in's (95% patient-bootstrap interval 0.62–1.07); median signed bias changes from -53% to $+6\%$. The error reduction is not statistically significant. This narrow mask-distance sensitivity does not resolve optical spillover or segmentation artifacts (Appendix Table 20).

H&E-derived geometry. A single-seed control segments H&E nuclei with InstanSeg [27], adds $1.95\,\mu$m rims, re-extracts features and refits all models. Across 40 patients and 79,600 cell proxies, clustering MSE is 0.70 of its own plug-in's (95% interval 0.62–0.89), versus 0.72 for the direct graph. Overall calibrated interval score is 12.2% below the direct model's at 90.6% coverage for both. These proxy targets differ from the primary whole-cell targets; optical artifacts remain unresolved (Appendix Table 24).

7 Discussion

LIGATURE trains a shared molecular representation against spatial analyses: the same joint draws support scored statistics and additional queries without refitting the generator. Compared with the matched direct graph, its supported advantage is a lower calibrated overall interval score, an exploratory ORION result driven by cross-marker forecasts that also holds against simple post hoc baselines; an overall point-prediction advantage is not demonstrated. One constant per statistic removes cohort bias and much of the clustering gain, so attenuation alone does not justify a generator. For a fixed query, direct prediction can be cheaper and more accurate, as on the new Moran and Geary queries. Ablations and independent-cell controls support statistic-level scoring, cross-cell dependence and shared patient variation; the identification analysis characterizes dependence left unspecified by the scores.

Limitations. Validation concerns selected statistics, not the full molecular joint law; samples are model realizations. Both cohorts (40 and 19 patients) were examined during development, and lung models are refitted within cohort. The mask-exclusion check does not correct spillover, so some measured clustering may reflect assay artifacts. Statistics use observed crop pairs without edge correction.

Percentile gates are phenotype proxies, and Gaussian residuals misfit marker tails; a copula helps gated queries. Calibration needs labeled patients. Clustering gains depend on intensity scale, and cross-platform stains understate their in-domain accuracy. All six stains are deterministic; we did not benchmark diffusion or adversarial translators that sample images directly. Anchor-marker conditioning failed (Table 11), consistent with Proposition 4(ii).

Code availability. `https://github.com/ark248/LIGATURE`; core release: `release/ligature`.

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